

Q1 - 24 January - Shift 1

Match List I with List II

LIST I		LIST II	
A.	Planck's constant (h)	I.	$[M^1 L^2 T^{-2}]$
B.	Stopping potential (Vs)	II.	$[M^1 L^1 T^{-1}]$
C.	Work function (ϕ)	III.	$[M^1 L^2 T^{-1}]$
D.	Momentum (p)	IV.	$[M^1 L^2 T^{-3} A^{-1}]$

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-IV, C-III, D-I

(4) A-I, B-III, C-IV, D-II

Space for your notes:

Q2 - 24 January - Shift 2

The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as :

$\nu = r^a \rho^b s^c$. The values of a , b and c respectively are

(1) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (2) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (3) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ (4) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Q3 - 25 January - Shift 1

Match List I with List II

List – I		List - II	
A	Surface tension	I.	$\text{Kg m}^{-1} \text{s}^{-1}$
B	Pressure	II.	Kg ms^{-1}
C	Viscosity	III.	$\text{Kg m}^{-1} \text{s}^{-2}$
D	Impulse	IV.	Kg s^{-2}

Choose the correct answer from the options given below :

- (1) A-IV, B-III, C- II, D- I
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-I, C-III, D-IV

Space for your notes:

Q4 - 25 January - Shift 2

List		List II	
A.	Young's Modulus (Y)	I.	$[\text{M L}^{-1} \text{T}^{-1}]$
B.	Co-efficient of Viscosity (η)	II.	$[\text{M L}^2 \text{T}^{-1}]$
C.	Planck's Constant (h)	III.	$[\text{M L}^{-1} \text{T}^{-2}]$
D.	Work Function (ϕ)	IV.	$[\text{M L}^2 \text{T}^{-2}]$

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-I, B-III, C-IV, D-II
- (4) A-I, B-II, C-III, D-IV

Space for your notes:

Q5 - 29 January - Shift 1

Match List I with List II :

List-I (Physical Quantity)		List-II (Dimensional Formula)	
A	Pressure gradient	I	$[M^0 L^2 T^{-2}]$
B	Energy density	II	$[M^1 L^{-1} T^{-2}]$
C	Electric Field	III	$[M^1 L^{-2} T^{-2}]$
D	Latent heat	IV	$[M^1 L^1 T^{-3} A^{-1}]$

Space for your notes:

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-III, C-I, D-IV

Q6 - 29 January - Shift 2

The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than $(0, 0)$, then find out the correct dimensions of A and B in a new

equation : $(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$.

The dimensions of t is given as $[T^{-1}]$.

- (1) $A = [L^{-1} T]$, $B = [LT^{-1}]$
- (2) $A = [LT]$, $B = [L^{-1} T^{-1}]$
- (3) $A = [L^{-1} T^{-1}]$, $B = [LT^{-1}]$
- (4) $A = [L^{-1} T^{-1}]$, $B = [LT]$

Q7 - 30 January - Shift 1

Electric field in a certain region is given by

$$\vec{E} = \left(\frac{A}{x^2} \hat{i} + \frac{B}{y^3} \hat{j} \right). \text{ The SI unit of A and B are :}$$

- (1) $\text{Nm}^3\text{C}^{-1}; \text{Nm}^2\text{C}^{-1}$ (2) $\text{Nm}^2\text{C}^{-1}; \text{Nm}^3\text{C}^{-1}$
 (3) $\text{Nm}^3\text{C}; \text{Nm}^2\text{C}$ (4) $\text{Nm}^2\text{C}; \text{Nm}^3\text{C}$

Space for your notes:

Q8 - 30 January - Shift 2

Match List I with List II.

List I		List II	
A	Torque	I.	$\text{kg m}^{-1} \text{s}^{-2}$
B	Energy density	II.	kg ms^{-1}
C	Pressure gradient	III.	$\text{kg m}^{-2} \text{s}^{-2}$
D	Impulse	IV.	$\text{kg m}^2 \text{s}^{-2}$

Space for your notes:

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
 (2) A-I, B-IV, C-III, D-II
 (3) A-IV, B-I, C-II, D-III
 (4) A-IV, B-I, C-III, D-II

Q9 - 31 January - Shift 1

If R , X_L and X_C represent resistance, inductive reactance and capacitive reactance. Then which of the following is dimensionless:

- (1) $R X_L X_C$ (2) $\frac{R}{\sqrt{X_L X_C}}$
 (3) $\frac{R}{X_L X_C}$ (4) $R \frac{X_L}{X_C}$

Space for your notes:

Q10 - 31 January - Shift 2

Match List-I with List-II.

	List-I		List-II
A.	Angular momentum	I	$[ML^2T^{-2}]$
B	Torque	II	$[ML^{-2}T^{-2}]$
C	Stress	III	$[ML^2T^{-1}]$
D	Pressure gradient	IV	$[ML^{-1}T^{-2}]$

Choose the correct answer from the options given below:

- (1) A-I, B-IV, C-III, D-II
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-II, C-I, D-III

Space for your notes:

Q11 - 01 February - Shift 1

$\left(P + \frac{a}{V^2}\right)(V - b) = RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a , b , R are the constants. The physical quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will be :

- (1) Bulk modulus
- (2) Modulus of rigidity
- (3) Compressibility
- (4) Energy density

Space for your notes:

Q12 - 01 February - Shift 2

If the velocity of light c , universal gravitational constant G and planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is:

Space for your notes:

(1) $\left[h^{\frac{1}{2}} c^{-\frac{1}{2}} G^1 \right]$ (2) $\left[h^1 c^1 G^{-1} \right]$

(3) $\left[h^{-\frac{1}{2}} c^{\frac{1}{2}} G^{\frac{1}{2}} \right]$ (4) $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^{-\frac{1}{2}} \right]$

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2)

Q2 (1)

Q3 (2)

Q4 (2)

Q5 (3)

Q6 (2)

Q7 (2)

Q8 (4)

Q9 (2)

Q10 (2)

Q11 (3)

Q12 (4)

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Q1 (2)

(A) Planck's constant

$$h\nu = E$$

$$h = \frac{E}{\nu} = \frac{M^1 L^2 T^{-2}}{T^{-1}} = M^1 L^2 T^{-1} \quad (\text{III})$$

(B) $E = qV$

$$V = \frac{E}{q} = \frac{M^1 L^2 T^{-2}}{A^1 T^1} = M^1 L^2 T^{-3} A^{-1} \quad (\text{IV})$$

(C) ϕ (work function) = energy

$$= M^1 L^2 T^{-2} \quad (\text{I})$$

(D) Momentum (p) = $F \cdot t$

$$= M^1 L^1 T^{-2} T^1$$

$$= M^1 L^1 T^{-1} \quad (\text{II})$$

Q2 (1)

$$[T^{-1}] = [L^1]^a [M^1 L^{-3}]^b \left[\frac{MLT^{-2}}{L} \right]^c$$

$$\Rightarrow T^{-1} = M^{b+c} \cdot L^{a-3b} \cdot T^{-2c}$$

$$c = \frac{1}{2}, \quad b = -\frac{1}{2}, \quad a - 3b = 0$$

$$a + \frac{3}{2} = 0 \Rightarrow a = -\frac{3}{2}$$

Q3 (2)

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$$(A) \text{ Surface Tension} = \frac{F}{\ell} = \frac{MLT^{-2}}{L} = ML^{-1}T^{-2}$$

$$= Kgs^{-2} \text{ (IV)}$$

$$(B) \text{ Pressure} = \frac{F}{A} = \frac{MLT^{-2}}{L^2}$$

$$= kg \, m^{-1}s^{-2} \text{ (III)}$$

$$(C) \text{ Viscosity} = \frac{F}{A \left(\frac{dV}{dz} \right)} = \frac{MLT^{-2}}{L^2 \left(\frac{LT^{-1}}{L} \right)}$$

$$= ML^{-1}T^{-1} = kg \, m^{-1}s^{-1} \text{ (I)}$$

$$(D) \text{ Impulse} = \int F dt = MLT^{-2} \times T$$

$$= MLT^{-1} = Kgs^{-1} \text{ (II)}$$

Q4 (2)

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta \ell / \ell} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$$

$$F = 6\pi\eta rv \Rightarrow \eta = \frac{F}{6\pi rv}$$

$$[\eta] = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1}T^{-1}]$$

$$E = h\nu \Rightarrow h = \frac{E}{\nu} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Work function has same dimension as that of energy, so $[\phi] = [ML^2T^{-2}]$

Q5 (3)

$$\text{Pressure gradient} = \frac{dp}{dx} = \frac{[ML^{-1}T^{-2}]}{[L]}$$

$$= [M^1 L^{-2} T^{-2}]$$

$$\text{Energy density} = \frac{\text{energy}}{\text{volume}} = \frac{[ML^2T^{-2}]}{[L^3]}$$

$$= [M^1 L^{-1} T^{-2}]$$

$$\text{Electric field} = \frac{\text{Force}}{\text{charge}} = \frac{[MLT^{-2}]}{[A.T]}$$

$$= [M^1 L^1 T^{-3} A^{-1}]$$

$$\text{Latent heat} = \frac{\text{heat}}{\text{mass}} = \frac{[ML^2T^{-2}]}{[M]}$$

$$= [M^0 L^2 T^{-2}]$$

Q6 (2)

$$(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$$

$$[At] = A \times \frac{1}{T} = L$$

$$\therefore [A] = T^1 L^1$$

$$\frac{t}{B} \text{ is in meters}$$

$$\therefore \frac{1}{T[B]} = L$$

$$\therefore [B] = T^{-1} L^{-1}$$

$$\therefore \text{Correct Ans. (2)}$$

Q7 (2)

$$\vec{E} = \frac{A}{x^2} \hat{i} + \frac{B}{y^3} \hat{j}$$

$$\left[\frac{A}{x^2}\right] = NC^{-1} \Rightarrow [A] = Nm^2C^{-1}$$

Q8 (4)

Solution not Available

Q9 (2)

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All three have same dimension therefore $\frac{R}{\sqrt{X_L X_C}}$ is dimensionless.

Option 2

Q10 (2)

No Solution

Q11 (3)

$$[b] = [V]$$

$$\left[\frac{a}{b^2} \right] = [P] \quad \therefore \left[\frac{b^2}{a} \right] = \frac{1}{[P]} = \frac{1}{[B]} = [K]$$

Q12 (4)

Say dimensional formale of mass is $H^x C^y G^z$

$$M^1 = (ML^2T^{-1})^x (LT^{-1}) (M^{-1}L^3T^{-2})^z$$

$$M^1 L^0 T^0 = M^{x-z} L^{2x+y+3z} T^{-x-y-2z}$$

on comparing both side

$$x - z = 1$$

$$2x + y + 3z = 0$$

$$-x - y - 2z = 0$$

On solving above equations we get

$$x = \frac{1}{2} \quad y = \frac{1}{2} \quad z = \frac{-1}{2}$$